

FORM 2 THE PATENT ACT 1970 (39 OF 1970) & The Patent Rules, 2003 COMPLETE SPECIFICATION (See Section 10 and Rule 13)	
TITLE OF THE INVENTION- A MULTIMODAL HEALTH MONITORING SYSTEM VIA SMART MIRROR AND WEARABLE SYNCHRONIZATION	
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2. PREAMBLE OF THE DESCRIPTION	
Complete Specification The following specification describes the invention	
3. DESCRIPTION (Description shall start from next page)	
4. CLAIMS (claims shall start from separate page)	
5. DATE AND SIGNATURE (TO GIVEN AT THE END OF LAST PAGE OF SPECIFICATION)	
6. ABSTRACT OF THE INVENTION (to be given along with the complete specification on separate page)	

FIELD OF THE INVENTION

The present invention relates to a multimodal health monitoring system via smart mirror and wearable synchronization. More specifically, a multimodal health monitoring system via smart mirror and wearable synchronization for enhanced
5 interaction and health monitoring.

BACKGROUND OF THE INVENTION

The global emphasis on preventive healthcare and early detection of medical conditions has created a growing demand for non-intrusive, real-time health monitoring solutions that can be seamlessly integrated into daily routines.
10 Traditional medical diagnostics often require specialized equipment, trained personnel and periodic appointments, which may delay the identification of emerging health conditions and reduce patient compliance. Wearable health devices such as smartwatches and fitness bands have gained popularity in recent years by offering continuous physiological monitoring (e.g., heart rate, activity levels, sleep
15 quality). However, these devices often suffer from user fatigue due to the need for constant wear, limited sensing capabilities, and inconsistent user engagement.

Smart home technologies have emerged as a potential solution to integrate health monitoring into users' environments, but existing devices generally lack intuitive interfaces and passive engagement mechanisms. Meanwhile, conventional mirrors
20 remain a staple in personal spaces such as bathrooms, bedrooms, and fitness areas, yet they are underutilized as platforms for delivering digital health insights. Furthermore, existing smart mirrors that incorporate health features often rely on manual data entry or peripheral sensors, reducing their practicality and scope for comprehensive health assessment.

25 US10,845,511B2 discloses in one example, a smart mirror. The smart mirror includes a display adjacent one side of a half-mirror, sensors to acquire biometric information of a user facing an opposite side of the half-mirror and a controller coupled to the sensors and the display. The controller determines from the biometric information a focal distance of the user. If the focal distance corresponds to the

display, the controller presents a user interface on the smart mirror in a highly-visible manner and a reflection of the user in a less-visible manner. If the focal distance corresponds to the reflection of the user, the controller presents the user interface on the smart mirror in a less-visible manner and the reflection of the user
5 in a highly-visible manner.

US10,194,862B2 discloses a wearable device comprising a display dial configured to display various health parameters, a wristband assisting the device to wear on wrist, an eject-able tray comprising a micro-chip, a first spring coupled to the eject-able tray, at least one latch provided with a second spring to hold the eject-able tray
10 within the device by compressing the first spring, and a health monitoring unit provided with multiple sensors to determine various health parameters, wherein compression of the second spring results in the latch to release the eject-able tray which in turn relaxes the compressed first spring to eject the micro-chip outside the device for collecting blood samples. The microchip comprises at least one micro-
15 needle and an enzyme test strip for collecting and analyzing the blood samples. The health monitoring unit comprises at least three conductive sense pads which are collectively operable to provide electrocardiograph (ECG) information.

Several inventions have previously explored the domains of smart mirrors and wearable health monitoring devices, but each in a limited or isolated manner. The
20 existing technology does not integrate any physiological monitoring or connect to external health devices like smartwatches. Furthermore, the existing invention focuses only on the wearable and lacks integration with any ambient interface like a smart mirror. These existing systems are portable and multifunctional but do not include any user-facing visual interface like a mirror for immediate, ambient
25 feedback. These systems generally do not read internal health parameters like pulse, temperature or synchronize with a wearable device in real time.

Therefore, there is a need for a multimodal health monitoring system via smart mirror and wearable synchronization that passively and continuously monitors multiple health indicators such as heart rate, respiratory rate, skin condition, body

posture, and emotional state, while maintaining an intuitive, non-disruptive user experience.

OBJECTIVE OF THE INVENTION

The main objective of the present invention is to develop an artificial intelligence
5 based multimodal health monitoring system.

Yet another objective of the present invention is to synchronize a wearable smartwatch and a smart mirror into a single interactive platform for health monitoring.

Yet another objective of the present invention is to develop facial based wellness
10 detection using image processing.

Yet another objective of the present invention is to monitor health statistics data in real time, offering personalized views and suggestions.

Yet another objective of the present invention is to implement hands free operation with visual feedback by integrating non-touch, camera and sensor-based
15 interactions allowing user to receive health updates.

SUMMARY OF THE INVENTION

The present invention relates to a multimodal health monitoring system via smart mirror and wearable synchronization. The present invention includes a smart mirror unit, a smartwatch and a power adapter. The smart mirror unit includes a two-way
20 mirror glass, a display screen, a camera, a motion sensor, a temperature sensor, a microphone, a speaker, a wireless module and a computing unit. The two-way mirror glass enables screen to be visible behind reflective surface, functioning both as mirror and display. The display screen is positioned behind the two-way mirror glass and configured to display time, notifications, health statistics data, daily health
25 reminders, weather updates, calendar and yoga instructions. The display screen is selected from LCD or LED. The camera is used for facial recognition and monitoring yoga postures. The camera utilizes OpenCV or Media Pipe for facial recognition and monitoring yoga postures. The motion sensor enables touchless

control via hand gestures for menu navigation and commands. The motion sensor is selected from ultrasonic sensor and infrared sensor. The temperature sensor is non-contact sensor and measures temperature regularly in real-time. Herein, the microphone and the speaker facilitate voice-based interactions between user and the smart mirror unit. The microphone is selected from micro electro mechanical system. The computing unit is connected to the two-way mirror glass, the display screen, the camera, the motion sensor, the temperature sensor, the microphone, the speaker and the wireless module. The computing unit includes a memory unit and a processing unit. The computing unit is selected from Raspberry Pi series (Raspberry Pi 4, Pi 5), Beagle Bone Black, Jetson Nano, Latte Panda, and ODROID. The memory unit stores data of user's health records, logs and backup data, computer readable instruction and artificial intelligence model. The processing unit pre-process input health data of user and generate personalized health suggestions and reminders to be displayed on the display screen. Herein, the processing unit runs artificial intelligence models stored in the memory unit, configured for health data processing. The smartwatch is connected wirelessly to the smart mirror unit through the wireless module and configured to monitor health statistics and physical activity of user and send monitored data to the smart mirror unit. Herein, the smartwatch measures heart rate, body temperature, step count and sleep data of user. The power adapter is connected to the smart mirror unit and provides constant power supply. Herein, the computing unit pre-process the input data receive from the two-way mirror glass, the camera, the motion sensor, the temperature sensor and the smartwatch, input-data processed from the computing unit is display on the display screen and communicated to user via audio through the microphone and speaker. The system monitors health statistics data in real-time, offering personalized views and suggestions for seamless and continuous wellness monitoring. The system supports multi-user face recognition automatically and showing each user health statistics dashboard separately, making perfect for family use. Herein, artificial intelligence model is selected from natural language processing, machine learning and deep learning. The display screen uses graphical

user interface for display. The memory unit stores cloud integration and remote artificial intelligence modelling for advance monitoring.

The main advantage of the present invention is that the system monitors health statistics data in real-time, offering personalized views and suggestions.

- 5 Yet another advantage of the present invention is that the system synchronize a wearable smartwatch and a smart mirror into a single interactive platform for health monitoring.

- Yet another advantage of the present invention is that the system implement hands free operation with visual feedback by integrating non-touch, camera and sensor-based interactions allowing user to receive health updates.
- 10

Yet another advantage of the present invention is that the system supports multi-user face recognition automatically and showing each user health dashboard separately, making perfect for family use.

- Yet another advantage of the present invention is that the system has compact and elegant design, making it suitable for moder smart homes.
- 15

Further objectives, advantages, and features of the present invention will become apparent from the detailed description provided herein below, in which various embodiments of the disclosed invention are illustrated by way of example.

BRIEF DESCRIPTION OF THE DRAWINGS

- 20 The accompanying drawings are incorporated in and constitute a part of this specification to provide a further understanding of the invention. The drawings illustrate one embodiment of the invention and together with the description, serve to explain the principles of the invention.

- Fig. 1 illustrates block diagram of a multimodal health monitoring system via smart mirror and wearable synchronization.
- 25

Fig. 2 illustrates three-dimensional prototype of a multimodal health monitoring system via smart mirror and wearable synchronization.

Fig. 3 illustrates flowchart/workflow of a multimodal health monitoring system via smart mirror and wearable synchronization.

DETAILED DESCRIPTION OF THE INVENTION

Definition

5 The terms “a” or “an”, as used herein, are defined as one or as more than one. The term “plurality”, as used herein, is defined as two or more than two. The term “another”, as used herein, is defined as at least a second or more. The terms “including” and/or “having”, as used herein, are defined as comprising (i.e., open language). The term “coupled”, as used herein, is defined as connected, although
10 not necessarily directly, and not necessarily mechanically.

The term “comprising” is not intended to limit inventions to only claiming the present invention with such comprising language. Any invention using the term comprising could be separated into one or more claims using “consisting” or “consisting of” claim language and is so intended. The term “comprising” is used
15 interchangeably used by the terms “having” or “containing”.

Reference throughout this document to “one embodiment”, “certain embodiments”, “an embodiment”, “another embodiment”, and “yet another embodiment” or similar terms means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the
20 present invention. Thus, the appearances of such phrases or in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics are combined in any suitable manner in one or more embodiments without limitation.

The term “or” as used herein is to be interpreted as an inclusive or meaning any one
25 or any combination. Therefore, “A, B or C” means any of the following: “A; B; C; A and B; A and C; B and C; A, B and C”. An exception to this definition will occur only when a combination of elements, functions, steps, or acts are in some way inherently mutually exclusive.

As used herein, the term "one or more" generally refers to, but not limited to, singular as well as the plural form of the term.

The drawings featured in the figures are to illustrate certain convenient embodiments of the present invention and are not to be considered as a limitation to that. Term "means" preceding a present participle of operation indicates the desired function for which there is one or more embodiments, i.e., one or more methods, devices, or apparatuses for achieving the desired function and that one skilled in the art could select from these or their equivalent in view of the disclosure herein and use of the term "means" is not intended to be limiting disclosure herein and use of the term "means" is not intended to be limiting.

Fig. 1 illustrates block diagram of a multimodal health monitoring system via smart mirror and wearable synchronization. The system (100) includes a smart mirror unit (102), a smartwatch (126) and a power adapter (128). The smart mirror unit (102) includes a two-way mirror glass (104), a display screen (106), a camera (108), a motion sensor (110), a temperature sensor (112), a microphone (114), a speaker (116), a wireless module (118) and a computing unit (120). The computing unit (120) is connected to the two-way mirror glass (104), the display screen (106), the camera (108), the motion sensor (110), the temperature sensor (112), the microphone (114), the speaker (116) and the wireless module (118). The computing unit (120) includes a memory unit (122) and a processing unit (124). The smartwatch (126) is connected wirelessly to the smart mirror unit (102) through the wireless module (118) for monitoring health statistics and physical activity of user and send monitored data to the smart mirror unit (102). The power adapter (128) is connected to the smart mirror unit (102) and provides constant power supply.

Fig. 2 illustrates three-dimensional prototype of a multimodal health monitoring system via smart mirror and wearable synchronization. The system (100) represents the smart mirror unit (102), the smart mirror unit (102) having a camera (108). The camera (108) is used for facial recognition and monitoring yoga postures. The smartwatch (126) is connected wirelessly to the smart mirror unit (102) through the

wireless module (118) for monitoring health statistics and physical activity of user and send monitored data to the smart mirror unit (102).

Fig. 3 illustrates flowchart/workflow of a multimodal health monitoring system via smart mirror and wearable synchronization. The method includes sensing health statics data of user by the smartwatch (126), wherein, health statics data includes heart rate, step count, body temperature and sleep data. Communicating sensed health statics data to the computing unit (120) by the smartwatch (126), the processing unit (124) uses artificial intelligence model stored in the memory unit (122) for analyzing health statics data and generate report or suggestions. Capturing image or video of user standing in front of the smart mirror unit (102) in real-time by the camera (108) and captured image or video is communicated to the computing unit (120), the computing unit (120) recognizes captured image or video and analyses facial expressions of user and detecting emotional states of user. Monitoring yoga postures of user in real-time based on video data received from the camera (108) by the computing unit (120), determining by the computing unit (120) using artificial intelligence model, whether yoga posture of user conforms to predefined acceptable standards. If yoga posture is determined to be inappropriate, displaying alert on the display screen (106) associated with the system (100) prompting user to correct posture and suggesting healthy habits. Detecting hand gestures of user in proximity to the display screen (106) by the motion sensor (110), interpreting detected hand gestures for enabling non-contact navigation of menu and commands on the display screen (106) by the computing unit (120). Measuring body temperature of user by the temperature sensor (112). If body temperature is above threshold, then generating and displaying alert on the display screen (106) advising user to take precautionary actions. Sensed data from the smartwatch (126), the camera (108), the motion sensor (110), and the temperature sensor (112) is centrally processed by the computing unit (120) in real-time. The processing unit (124) uses artificial intelligence model stored in the memory unit (122) to analyze health statics data and generate personalized health suggestions and reminders. Displaying on the display screen (106) processed health statistics data and personalized health suggestions and reminders. Communicating via the microphone

(114) and the speaker (116) through audio messages or voice-based feedback to and from user.

The present invention relates to a multimodal health monitoring system via smart mirror and wearable synchronization. The present invention includes a smart mirror unit, a smartwatch and a power adapter. The smart mirror unit includes a two-way mirror glass, a display screen, a camera, a motion sensor, a temperature sensor, a microphone, a speaker, a wireless module and a computing unit. The two-way mirror glass enables screen to be visible behind reflective surface, functioning both as mirror and display. The display screen is positioned behind the two-way mirror glass and configured to display time, notifications, health statistics data, daily health reminders, weather updates, calendar and yoga instructions. In an embodiment, the display screen is selected from LCD or LED. The camera is used for facial recognition and monitoring yoga postures. In an embodiment, the camera utilizes OpenCV or MediaPipe for facial recognition and monitoring yoga postures. The motion sensor enables touchless control via hand gestures for menu navigation and commands. In an embodiment, the motion sensor is selected from ultrasonic sensor and infrared sensor. The temperature sensor is non-contact sensor and measures temperature regularly in real-time. Herein, the microphone and the speaker facilitate voice-based interactions between user and the smart mirror unit. In an embodiment, the microphone is selected from micro electro mechanical system. The computing unit is connected to the two-way mirror glass, the display screen, the camera, the motion sensor, the temperature sensor, the microphone, the speaker and the wireless module. The computing unit includes a memory unit and a processing unit. In an embodiment, the computing unit is selected from Raspberry Pi series (Raspberry Pi 4, Pi 5), Beagle Bone Black, Jetson Nano, Latte Panda, and ODROID. The memory unit stores data of user's health records, logs and backup data, computer readable instruction and artificial intelligence model. The processing unit pre-process input health data of user and generate personalized health suggestions and reminders to be displayed on the display screen. Herein, the processing unit runs artificial intelligence model stored in the memory unit, configured for health data processing. The smartwatch is connected wirelessly to

- the smart mirror unit through the wireless module, the smartwatch monitors health statistics and physical activity of user and send monitored data to the smart mirror unit. Herein, the smartwatch measures heart rate, body temperature, step count and sleep data of user. The power adapter is connected to the smart mirror unit and
- 5 provides constant power supply. In an embodiment, the computing unit pre-process input data receive from the two-way mirror glass, the camera, the motion sensor, the temperature sensor and the smartwatch, input-data processed from the computing unit is display on the display screen and communicated to user via audio through the microphone and speaker.
- 10 In an embodiment the system monitors health statistics data in real-time, offering personalized views and suggestions for seamless and continuous wellness monitoring. In an embodiment, the system supports multi-user face recognition automatically and showing each user health statistics dashboard separately, making perfect for family use.
- 15 In an embodiment, artificial intelligence model is selected from natural language processing, machine learning and deep learning.
- In another embodiment, the display screen uses graphical user interface for display. In an embodiment, the memory unit stores cloud integration and remote artificial intelligence modelling for advance monitoring.
- 20 In an embodiment, a method for the multimodal health monitoring system via smart mirror and wearable synchronization includes:
- sensing health statics data of user by the smartwatch, wherein, health statics data includes heart rate, step count, body temperature and sleep data,
- communicating sensed health statics data to the computing unit by the smartwatch,
- 25 the processing unit uses artificial intelligence model stored in the memory unit for analyzing health statics data and generate report or suggestions;
- capturing image or video of user standing in front of the smart mirror unit in real-time by the camera and captured image or video is communicated to the computing unit,

- the computing unit recognizes captured image or video and analyses facial expressions of user and detecting emotional states of user, herein emotional states of user determine whether user is happy, sad, angry or calm,
- 5 monitoring yoga postures of user in real-time based on video data received from the camera by the computing unit,
- captured image or video is converted to RGB form and process angle and check point of body for correct posture determination,
- determining by the computing unit using artificial intelligence model, whether yoga posture of user conforms to predefined acceptable standards,
- 10 if yoga posture is determined to be inappropriate, displaying alert on the display screen associated with the system prompting user to correct posture and suggesting healthy habits;
- detecting hand gestures of user in proximity to the display screen by the motion sensor,
- 15 interpreting detected hand gestures for enabling non-contact navigation of menu and commands on the display screen by the computing unit;
- measuring body temperature of user by the temperature sensor,
- determining whether measured body temperature exceeds predefined threshold value corresponding to normal human body temperature by the computing unit,
- 20 if body temperature is above threshold, then generating and displaying alert on the display screen advising user to take precautionary actions;
- sensed data from the smartwatch, the camera, the motion sensor, and the temperature sensor is centrally processed by the computing unit in real-time,
- the processing unit uses artificial intelligence model stored in the memory unit to
- 25 analyze health statics data and generate personalized health suggestions and reminders;

displaying on the display screen processed health statistics data and personalized health suggestions and reminders; and

communicating via the microphone and the speaker through audio messages or voice-based feedback to and from user.

- 5 The present invention relates to a multimodal health monitoring system via smart mirror and wearable synchronization. The present invention includes a smart mirror unit, a one or more smartwatches and a one or more power adapters. The smart mirror unit includes a one or more two-way mirror glasses, a one or more display screens, a one or more cameras, a one or more motion sensors, a one or more
- 10 temperature sensors, a one or more microphones, a one or more speakers, a one or more wireless modules and a computing unit. The one or more two-way mirror glasses enables screen to be visible behind reflective surface, functioning both as mirror and display. The one or more display screens are positioned behind the one or more two-way mirror glasses and configured to display time, notifications, health
- 15 statistics data, daily health reminders, weather updates, calendar and yoga instructions. In an embodiment, the one or more display screens are selected from LCD or LED. The one or more cameras are used for facial recognition and monitoring yoga postures. In an embodiment, the one or more cameras utilizes OpenCV or MediaPipe for facial recognition and monitoring yoga postures. The
- 20 one or more motion sensors enables touchless control via hand gestures for menu navigation and commands. In an embodiment, the one or more motion sensors are selected from ultrasonic sensor and infrared sensor. The one or more temperature sensors are non-contact sensor and measures temperature regularly in real-time. Herein, the one or more microphones and the one or more speakers facilitates voice-
- 25 based interactions between user and the smart mirror unit. In an embodiment, the one or more microphones are selected from micro electro mechanical system. The computing unit is connected to the one or more two-way mirror glasses, the one or more display screens, the one or more cameras, the one or more motion sensors, the one or more temperature sensors, the one or more microphones, the one or more
- 30 speakers and the one or more wireless modules. The computing unit includes a

memory unit and a processing unit. In an embodiment, the computing unit is selected from Raspberry Pi series (Raspberry Pi 4, Pi 5), Beagle Bone Black, Jetson Nano, Latte Panda, and ODROID. The memory unit stores data of user's health records, logs and backup data, computer readable instruction and artificial intelligence model. The processing unit pre-process input health data of user and generate personalized health suggestions and reminders to be displayed on the one or more display screens. Herein, the processing unit runs artificial intelligence models stored in the memory unit, configured for health data processing. The one or more smartwatches are connected wirelessly to the smart mirror unit through the one or more wireless modules, the one or more smartwatches monitor health statistics and physical activity of user and send monitored data to the smart mirror unit. Herein, the one or more smartwatches measures heart rate, body temperature, step count and sleep data of user. The one or more power adapters are connected to the smart mirror unit and provides constant power supply. In an embodiment, the computing unit pre-process the input data receive from the one or more two-way mirror glasses, the one or more cameras, the one or more motion sensors, the one or more temperature sensors and the one or more smartwatches, input-data processed from the computing unit is display on the one or more display screens and communicated to user via audio through the one or more microphones and one or more speakers.

In an embodiment the system monitors health statistics data in real-time, offering personalized views and suggestions for seamless and continuous wellness monitoring. In an embodiment, the system supports multi-user face recognition automatically and showing each user health statistics dashboard separately, making perfect for family use.

In an embodiment, artificial intelligence model is selected from natural language processing, machine learning and deep learning.

In another embodiment, the one or more display screens uses graphical user interface for display.

In an embodiment, the memory unit stores cloud integration and remote artificial intelligence modelling for advance monitoring.

In an embodiment, a method for the multimodal health monitoring system via smart mirror and wearable synchronization includes:

sensing health statics data of user by the one or more smartwatches, wherein, health statics data includes heart rate, step count, body temperature and sleep data,

5 communicating sensed health statics data to the computing unit by the one or more smartwatches,

the processing unit uses artificial intelligence model stored in the memory unit for analyzing health statics data and generate report or suggestions;

capturing image or video of user standing in front of the smart mirror unit in real-time by the one or more cameras and captured image or video is communicated to
10 the computing unit,

the computing unit recognizes captured image or video and analyses facial expressions of user and detecting emotional states of user, herein emotional states of user determine whether user is happy, sad, angry or calm,

15 monitoring yoga postures of user in real-time based on video data received from the one or more cameras by the computing unit,

captured image or video is converted to RGB form and process angle and check point of body for correct posture determination,

determining by the computing unit using artificial intelligence model, whether yoga
20 posture of user conforms to predefined acceptable standards,

if yoga posture is determined to be inappropriate, displaying alert on the one or more display screens associated with the system prompting user to correct posture and suggesting healthy habits;

detecting hand gestures of user in proximity to the one or more display screens by
25 the one or more motion sensors,

interpreting detected hand gestures for enabling non-contact navigation of menu and commands on the one or more display screens by the computing unit;

measuring body temperature of user by the one or more temperature sensors,
determining whether measured body temperature exceeds predefined threshold
value corresponding to normal human body temperature by the computing unit,
if body temperature is above threshold, then generating and displaying alert on the
5 one or more display screens advising user to take precautionary actions;
sensed data from the one or more smartwatches, the one or more cameras, the one
or more motion sensors, and the one or more temperature sensors are centrally
processed by the computing unit in real-time,
the processing unit uses artificial intelligence model stored in the memory unit to
10 analyze health statics data and generate personalized health suggestions and
reminders;
displaying on the one or more display screens processed health statistics data and
personalized health suggestions and reminders; and
communicating via the one or more microphones and the one or more speakers
15 through audio messages or voice-based feedback to and from user.

I/WE CLAIM

1. A Multimodal Health Monitoring System (100) via smart mirror and wearable synchronization, the system (100) comprising:

a smart mirror unit (102), the smart mirror unit (102) having

- 5 an at least one two-way mirror glass (104), the at least one two-way mirror glass (104) enables screen to be visible behind reflective surface, functioning both as mirror and display,
- an at least one display screen (106), the at least one display screen (106) is positioned behind the at least one two-way mirror glass (104), the at least one
- 10 display screen (106) display time, notifications, health statistics data, daily health reminders, weather updates, calendar and yoga instructions,
- an at least one camera (108), the at least one camera (108) is used for facial recognition and monitoring yoga postures,
- an at least one motion sensor (110), the at least one motion sensor (110)
- 15 enables touchless control via hand gestures for menu navigation and commands,
- an at least one temperature sensor (112), the at least one temperature sensor (112) is non-contact sensor and measures temperature regularly in real-time,
- an at least one microphone (114),
- 20 an at least one speaker (116),
- wherein, the at least one microphone (114) and the at least one speaker (116) facilitates voice-based interactions between user and the smart mirror unit (102),
- an at least one wireless module (118), and
- 25 a computing unit (120), the computing unit (120) is connected to the at least one two-way mirror glass (104), the at least one display screen (106), the at least one camera (108), the at least one motion sensor (110), the at least one temperature sensor (112), the at least one microphone (114), the at least one speaker (116) and the at least one wireless module (118), the computing unit
- 30 (120) having

a memory unit (122), the memory unit (122) stores data of user's health records, logs and backup data, computer readable instruction and artificial intelligence model, and

5 a processing unit (124), the processing unit (124) pre-process input health data of user and generate personalized health suggestions and reminders to be displayed on the at least one display screen (106), wherein, the processing unit (124) runs artificial intelligence models stored in the memory unit (122), configured for health data processing;

10 an at least one smartwatch (126), the at least one smartwatch (126) is connected wirelessly to the smart mirror unit (102) through the at least one wireless module (118), the at least one smartwatch (126) monitors health statistics and physical activity of user and send monitored data to the smart mirror unit (102); wherein, the at least one smartwatch (126) measures heart rate, body temperature, step count and sleep data of user;

15 an at least one power adapter (128), the at least one power adapter (128) is connected to the smart mirror unit (102) and provides constant power supply; wherein, the computing unit (120) pre-process the input data receive from the at least one two-way mirror glass (104), the at least one camera (108), the at least one motion sensor (110), the at least one temperature sensor (112) and the at least one

20 smartwatch (126), input-data processed from the computing unit (120) is display on the at least one display screen (106) and communicated to user via audio through the at least one microphone (114) and at least one speaker (116);

characterized in that, the system (100) monitors health statistics data in real-time, offering personalized views and suggestions for seamless and continuous wellness

25 monitoring,

characterized in that, the system (100) supports multi-user face recognition automatically and showing each user health statistics dashboard separately, making perfect for family use.

2. The system (100) as claimed in claim 1, wherein the computing unit (120) is selected from Raspberry Pi series (Raspberry Pi 4, Pi 5), Beagle Bone Black, Jetson Nano, Latte Panda, and ODROID.
3. The system (100) as claimed in claim 1, wherein artificial intelligence model is selected from natural language processing, machine learning and deep learning.
4. The system (100) as claimed in claim 1, wherein the at least one camera (108) utilizes OpenCV or Media Pipe for facial recognition and monitoring yoga postures.
5. The system (100) as claimed in claim 1, wherein the at least one display screen (106) is selected from LCD or LED.S
6. The system (100) as claimed in claim 1, wherein the at least one motion sensor (110) is selected from ultrasonic sensor and infrared sensor.
7. The system (100) as claimed in claim 1, wherein the at least one microphone (114) is selected from micro electro mechanical system.
8. The system (100) as claimed in claim 1, wherein the at least one display screen (106) uses graphical user interface for display.
9. The system (100) as claimed in claim 1, wherein the memory unit (122) stores cloud integration and remote artificial intelligence modelling for advance monitoring.
10. A method for the multimodal health monitoring system (100) via smart mirror and wearable synchronization, the method comprising:
 - sensing health statics data of user by the at least one smartwatch (126), wherein, health statics data comprising heart rate, step count, body temperature and sleep data;
 - communicating sensed health statics data to the computing unit (120) by the at least one smartwatch (126);
 - the processing unit (124) uses artificial intelligence model stored in the memory unit (122) for analyzing health statics data and generate report or suggestions;
 - capturing image or video of user standing in front of the smart mirror unit (102) in real-time by the at least one camera (108) and captured image or video is communicated to the computing unit (120);

the computing unit (120) recognizes captured image or video and analyses facial expressions of user and detecting emotional states of user, wherein emotional states of user determine whether user is happy, sad, angry or calm;

monitoring yoga postures of user in real-time based on video data received from the

5 at least one camera (108) by the computing unit (120);

captured image or video is converted to RGB form and process angle and check point of body for correct posture determination;

determining by the computing unit (120) using artificial intelligence model, whether yoga posture of user conforms to predefined acceptable standards;

10 if yoga posture is determined to be inappropriate, displaying alert on the at least one display screen (106) associated with the system (100) prompting user to correct posture and suggesting healthy habits;

detecting hand gestures of user in proximity to the at least one display screen (106) by the at least one motion sensor (110);

15 interpreting detected hand gestures for enabling non-contact navigation of menu and commands on the at least one display screen (106) by the computing unit (120);

measuring body temperature of user by the at least one temperature sensor (112),

determining whether measured body temperature exceeds predefined threshold value corresponding to normal human body temperature by the computing unit

20 (120);

if body temperature is above threshold, then generating and displaying alert on the at least one display screen (106) advising user to take precautionary actions;

sensed data from the at least one smartwatch (126), the at least one camera (108), the at least one motion sensor (110), and the at least one temperature sensor (112)

25 is centrally processed by the computing unit (120) in real-time;

the processing unit (124) uses artificial intelligence model stored in the memory unit (122) to analyze health statics data and generate personalized health suggestions and reminders;

5 displaying on the at least one display screen (106) processed health statistics data and personalized health suggestions and reminders; and

communicating via the at least one microphone (114) and the at least one speaker (116) through audio messages or voice-based feedback to and from user.



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ABSTRACT

A MULTIMODAL HEALTH MONITORING SYSTEM VIA SMART MIRROR AND WEARABLE SYNCHRONIZATION

The present invention relates to a multimodal health monitoring system(100) via smart mirror and wearable synchronization. The present invention includes a smart mirror unit(102), a smartwatch(126) and a power adapter(128). The smart mirror unit(102) includes a two-way mirror glass(104), a display screen(106), a camera(108), a motion sensor(110), a temperature sensor(112), a microphone(114), a speaker(116), a wireless module(118) and a computing unit(120). The computing unit(120) is connected to the two-way mirror glass(104), the display screen(106), the camera(108), the motion sensor(110), the temperature sensor(112), the microphone(114), the speaker(116) and the wireless module(118) and includes a memory unit(122) and a processing unit(124). The smartwatch(126) is connected wirelessly to the smart mirror unit(102) through the wireless module(118) for monitoring health statistics and physical activity of user and send monitored data to the smart mirror unit(102). The power adapter(128) is connected to the smart mirror unit(102) and provides constant power supply.

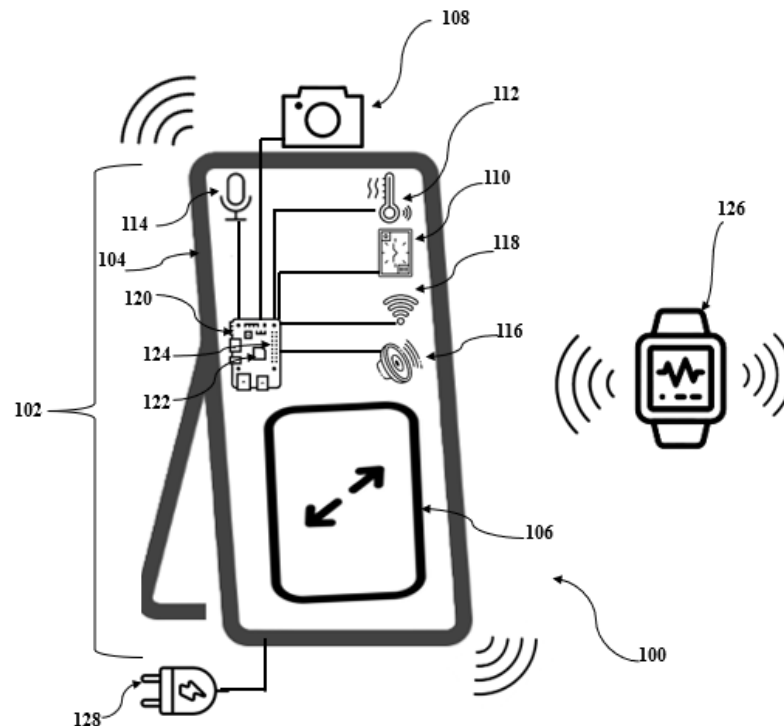


Fig. 1